

Ambuj Kumar Rai

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LinkedIn

GitHub

EDUCATION

Maulana Azad National Institute of Technology (MANIT), Bhopal

Jun 2024

Master of Computer Application (MCA)

CGPA: 8.49/10.0

TECHNICAL SKILLS

Languages: C/C++, Java, Python, JavaScript (ES6+), HTML5, CSS3

Web Development: React.js, Node.js, Express.js, REST APIs, MongoDB, MySQL

Tools & Technologies: Git/GitHub, VS Code, Postman, AWS, JDBC, Java Swing

Core Competencies: Full-Stack Web Development, REST API Design, Backend Engineering, Database Modeling, Software Development Lifecycle, **Data Structures & Algorithms**, **Algorithm Design & Analysis**, Operating Systems, Computer Networks, DBMS, SQL

PROFESSIONAL EXPERIENCE

GE Healthcare (via Tata Consultancy Services)

Jun 2024 – Present

Full Stack Developer

Kolkata, India

- Developed enterprise MERN applications managing 40K+ healthcare records with streamlined data workflows.
- Reduced page load time by 93.7% (8 mins to 30 secs) through HashMap caching and optimized database queries.
- Implemented advanced filtering and virtual scrolling, increasing large-table rendering speed by 4×.
- Built modular, reusable React components with robust error handling and over 85% test coverage.
- Designed scalable REST APIs using Node.js, Express.js, and MongoDB supporting high-throughput operations.
- Added auto-scroll functionality to improve user navigation across large datasets.
- Applied UI grouping techniques for structured data visualization and enhanced readability.

PROJECTS

Facebook Clone | MongoDB, Express.js, React.js, Node.js, JWT | 

Jun 2023 – Oct 2023

- Designed a full-stack social platform enabling secure login, posts, likes, and comments using JWT authentication.
- Refined MongoDB performance by 40% with advanced indexing and aggregation tuning.
- Crafted responsive React workflows and Express.js APIs for smooth CRUD operations.

ATM Software System | Java, Swing, MySQL, JDBC | 

May 2022 – Nov 2022

- Constructed a secure ATM simulator using Java, Swing, MySQL, and JDBC with robust validation.
- Developed withdrawal, deposit, and inquiry modules, reducing user error rate through structured validation logic.
- Delivered a streamlined Swing UI that improved navigation time by 30% through optimized component layouts.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

GeeksforGeeks | 

Coding Score: 2811

- Placed within the top five percent among global GFG users after solving 1256+ DSA problems.

LeetCode | 

Contest Rating: 1700

- Positioned in the top 12.7 percent globally on LeetCode with a contest rating of 1700.

HackerRank | 

Multiple Certifications

- Earned gold badges in C, C++, and Data Structures & Algorithms; achieved a silver badge in SQL

NIMCET National Entrance Exam

All India Rank: 422

- Achieved top one-percent standing among 50,000+ candidates in the NIMCET national exam.

CERTIFICATIONS & AWARDS

Certifications:  C++ (Udemy),  SQL & MySQL (Udemy),  Data Structures & Algorithms (GFG),  Problem Solving (HackerRank)

Awards:  Star of the Month,  Best Team Award,  Spot Award (2x),  TCS & GE Healthcare Appreciations